Entity Neutering

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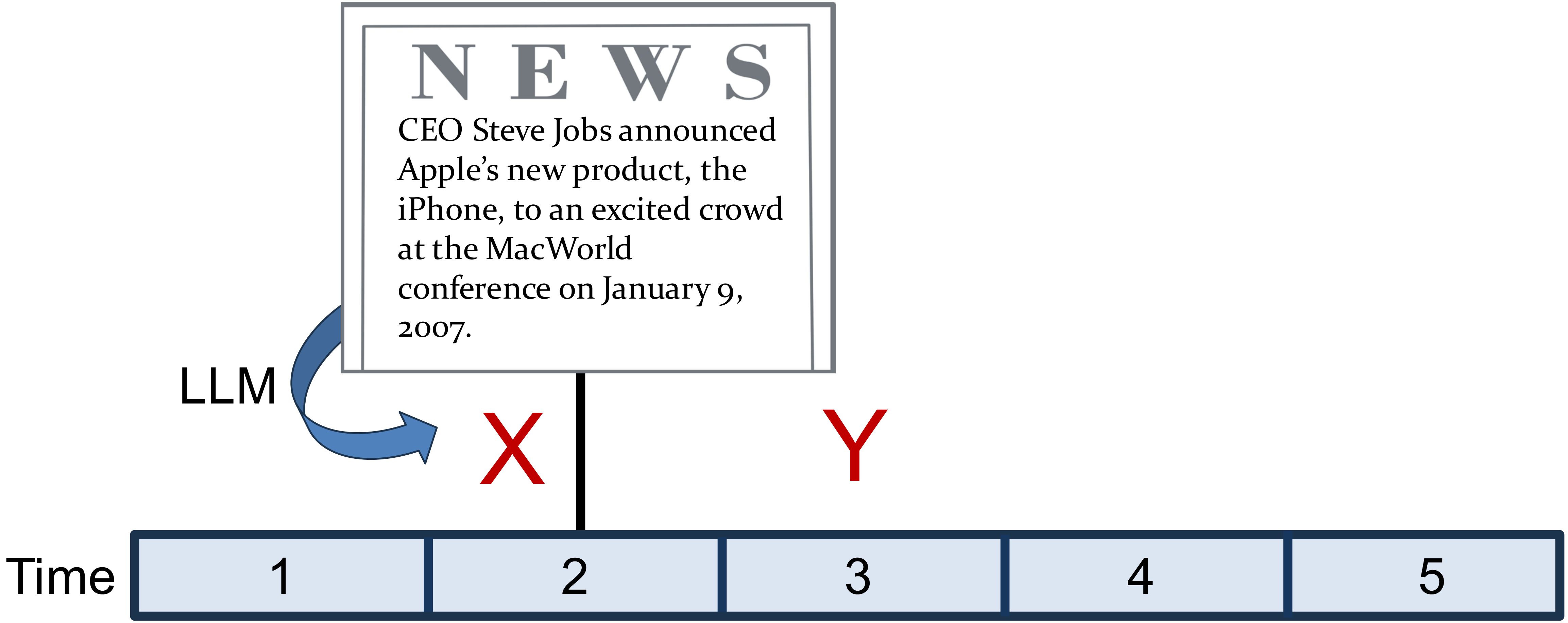
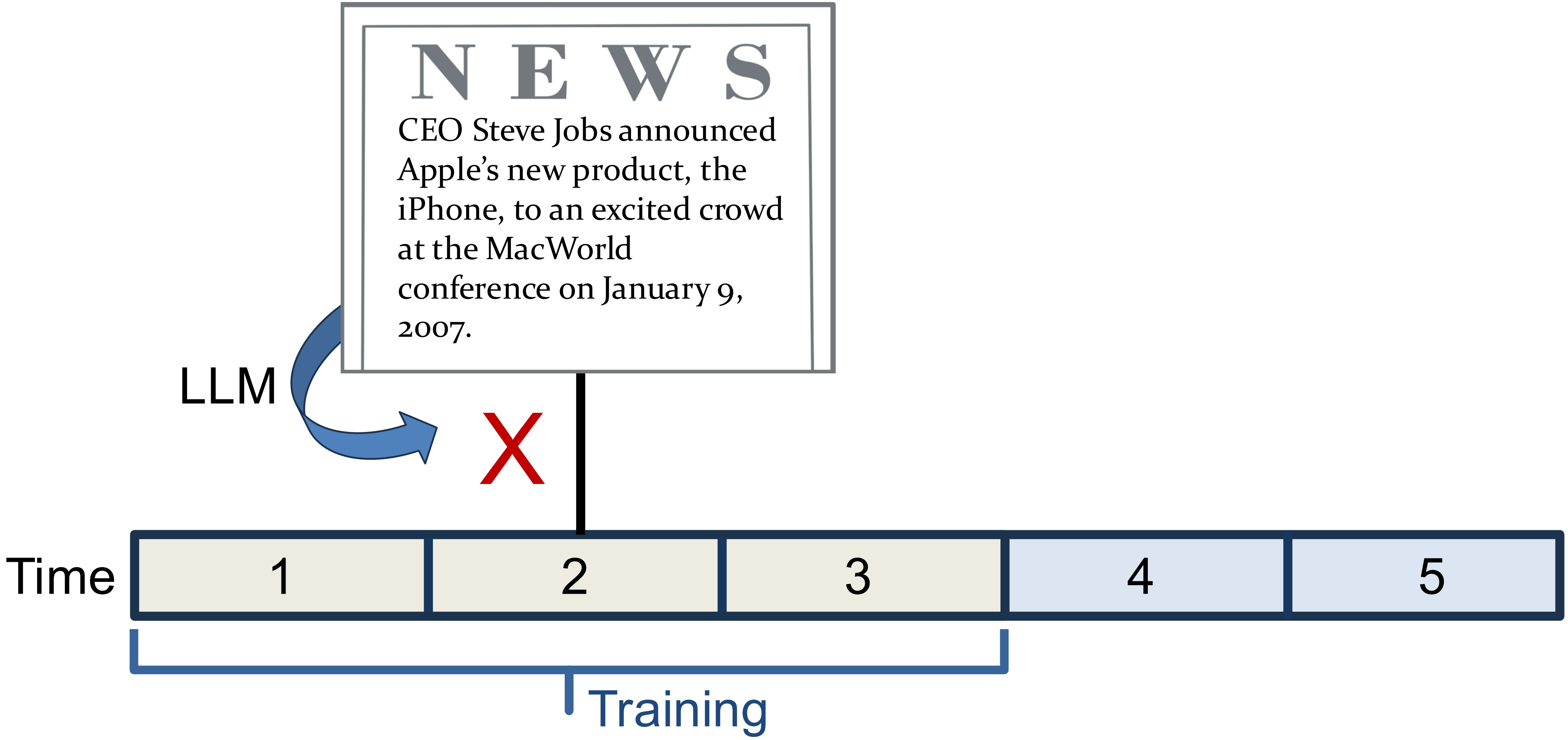
The Problem

- Explosive growth of Large Language Models (LLMs) in our papers
- Offer new opportunities to extract information from text for prediction purposes
- **However**, impressive predictions may be coming from look-ahead bias (Ludwig, Mullainathan, and Rambachan (2024), Sarkar and Vafa (2024) and Lopez-Lira, Tang and Zhu (2025))

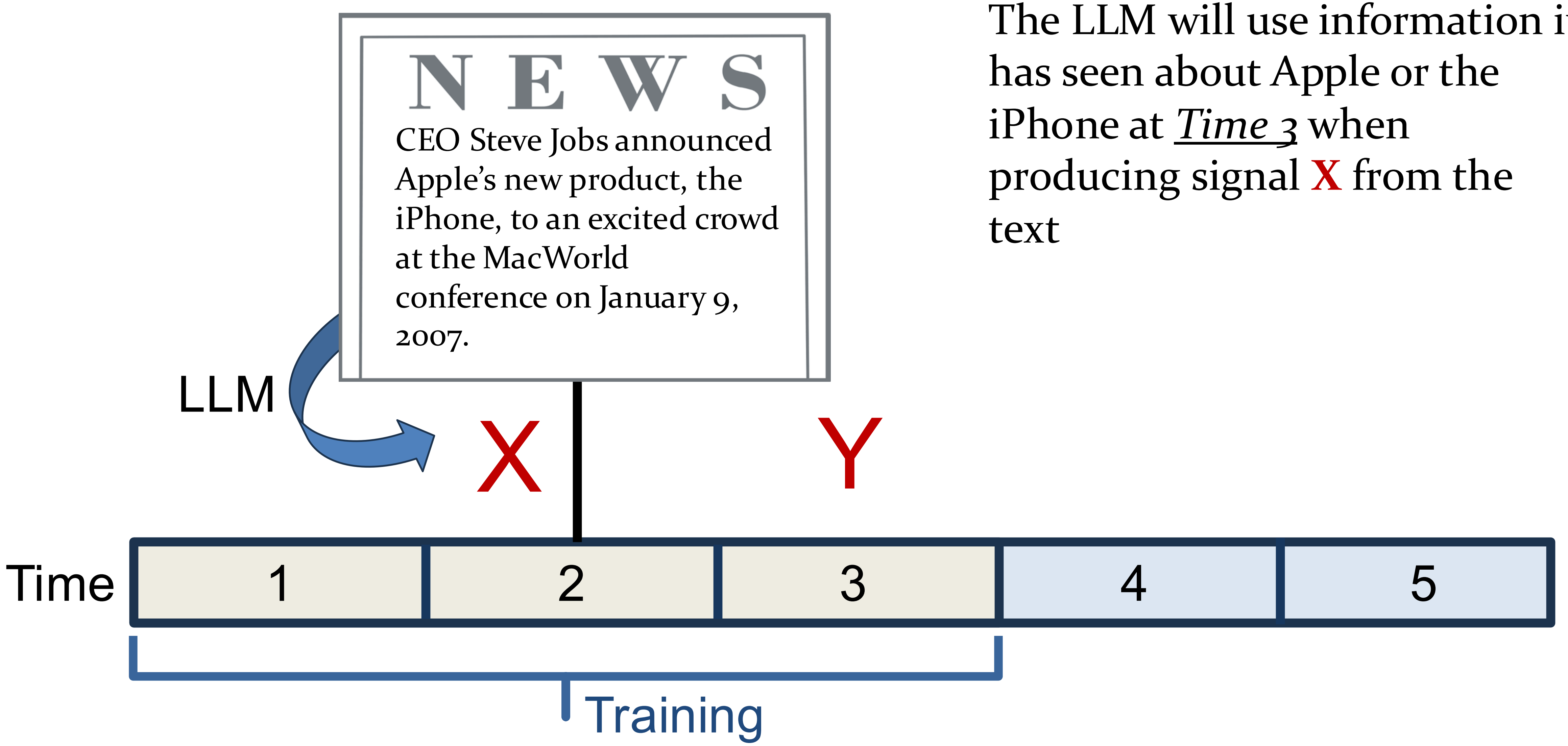
Stylized example: extract signal X, predict Y



The concern

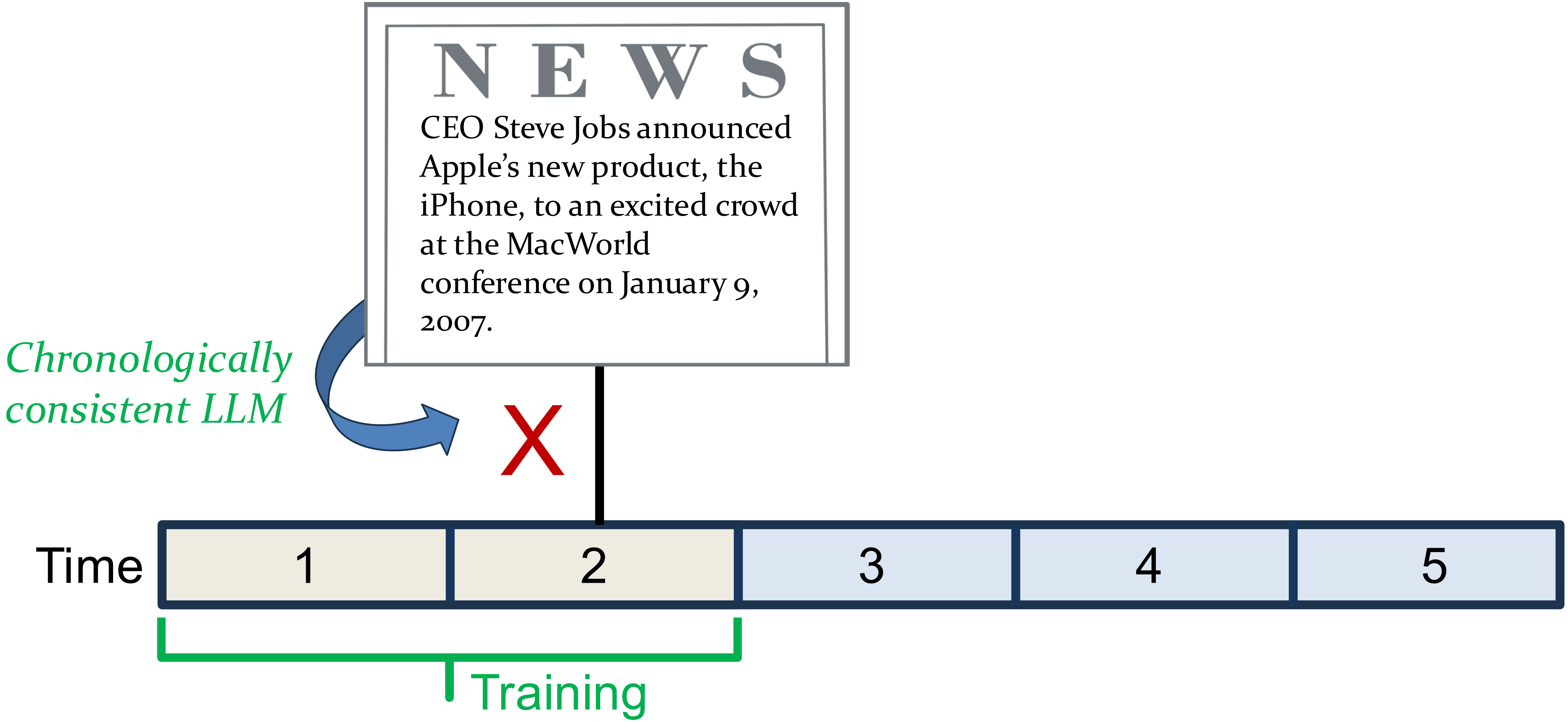

NEWS
CEO Steve Jobs announced Apple's new product, the iPhone, to an excited crowd at the MacWorld conference on January 9, 2007.
LLM
X
Time
1
2
3
4
5
Training

The concern



The LLM will use information it has seen about Apple or the iPhone at <u>*Time 3*</u> when producing signal X from the text

Solution #1: Chronological consistency
(Changing the model)



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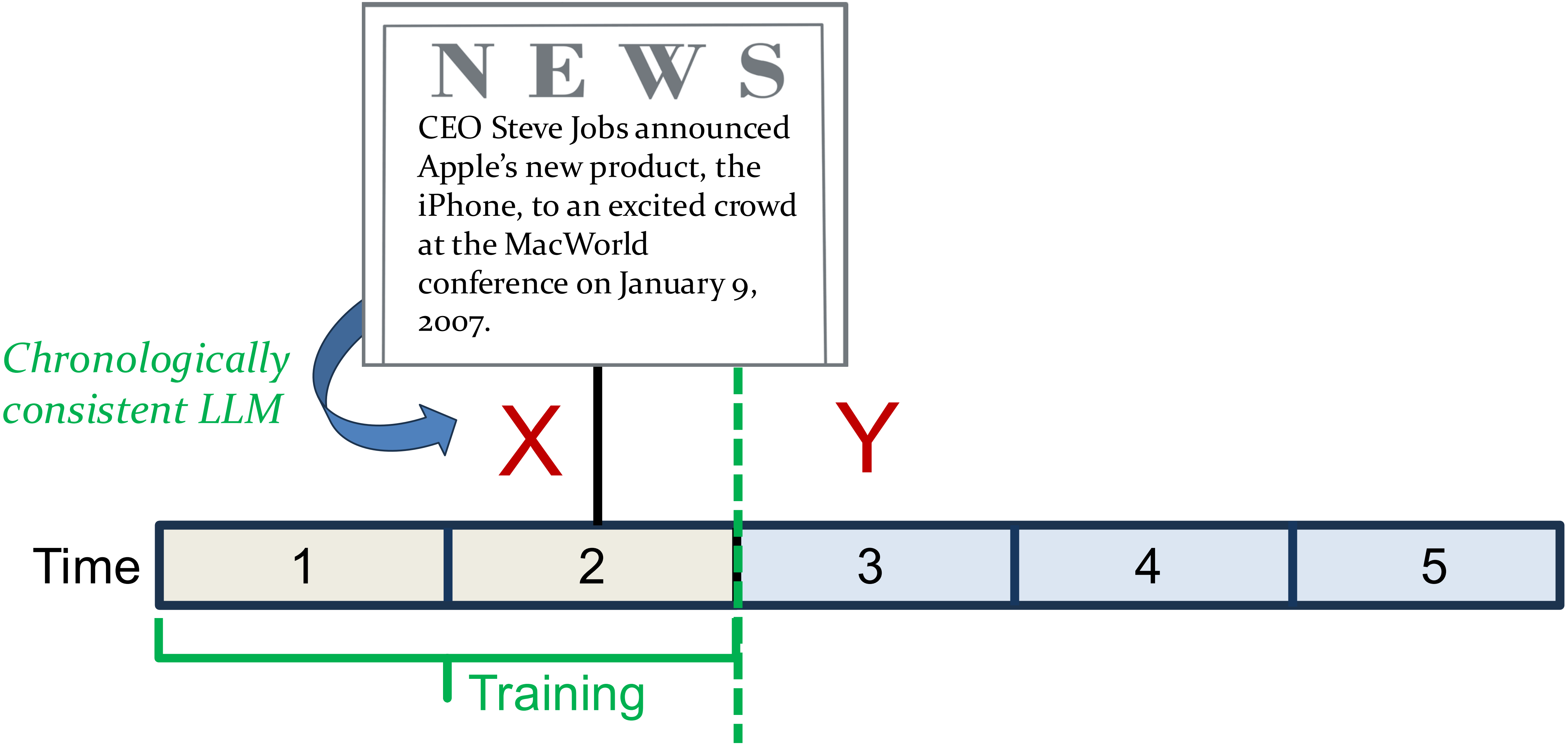

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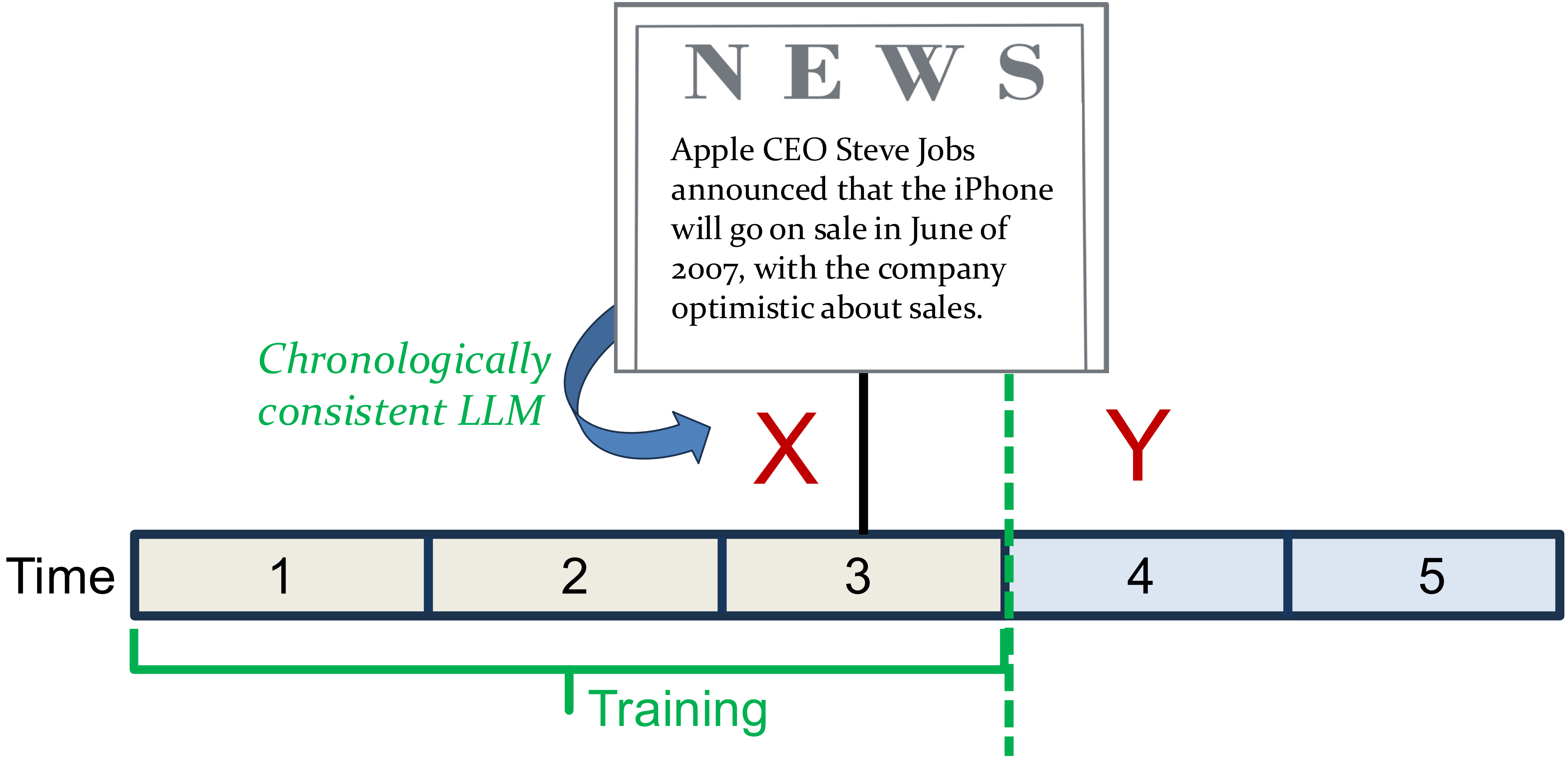

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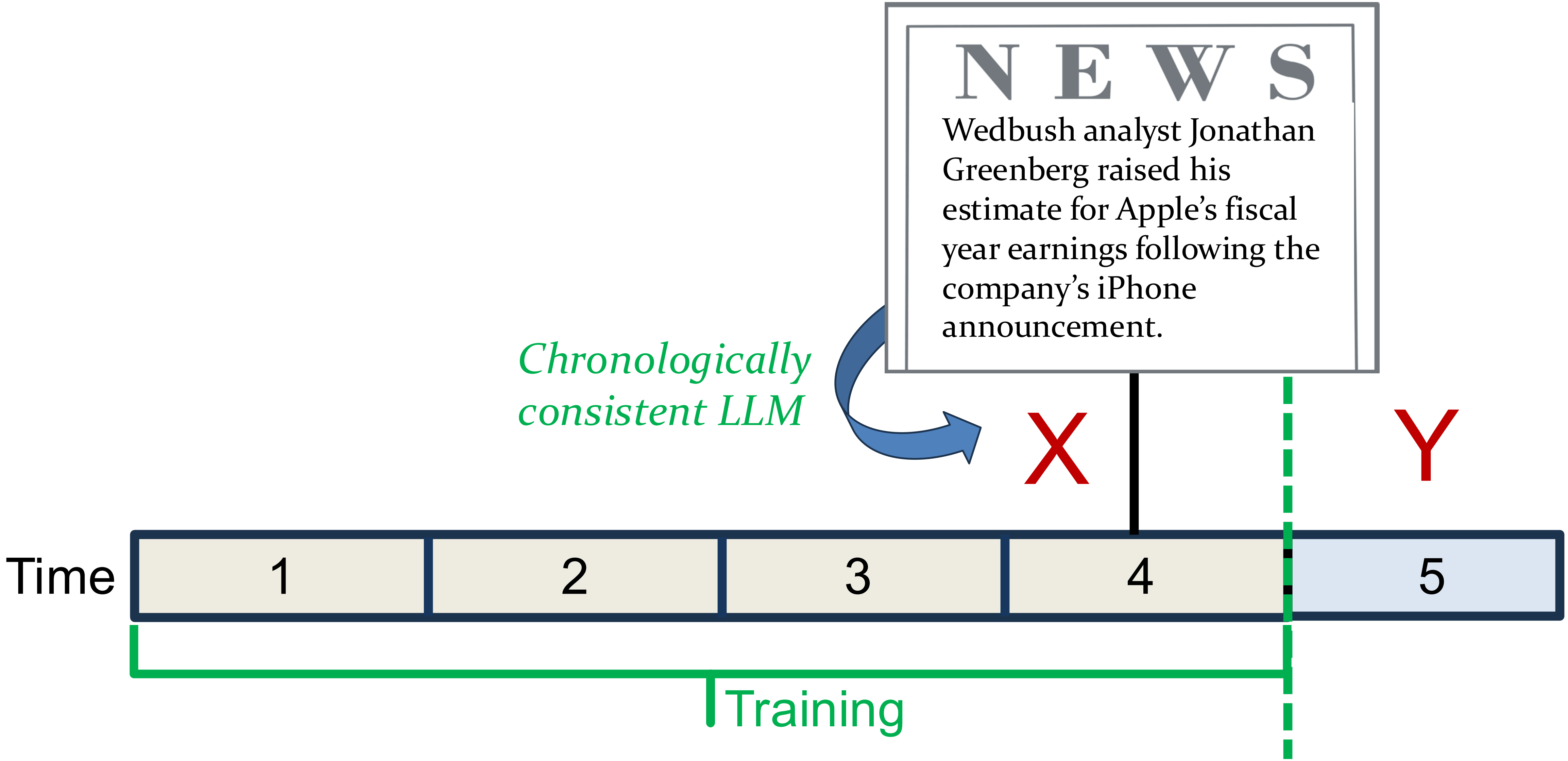


Chronological consistency in practice

<u>Changing the model</u>

- Sarkar (2024): time-indexed models using the American Stories corpus, a news dataset ending in 1963

- He, Lv, Manela and Wu (2025): yearly, point-in-time models starting in 1999

<u>Chronological consistency with frontier LLMs</u>

- Restrict sample to *after* LLM training cutoff date, or
- Check to see if paper's conclusions hold after cutoff
  - e.g., Jha, Qian, Weber and Yang 2024, Lopez-Lira and Tang 2024, Bybee 2025

Chronological consistency in practice

Advantages

- Look-ahead is impossible if the model never sees the future

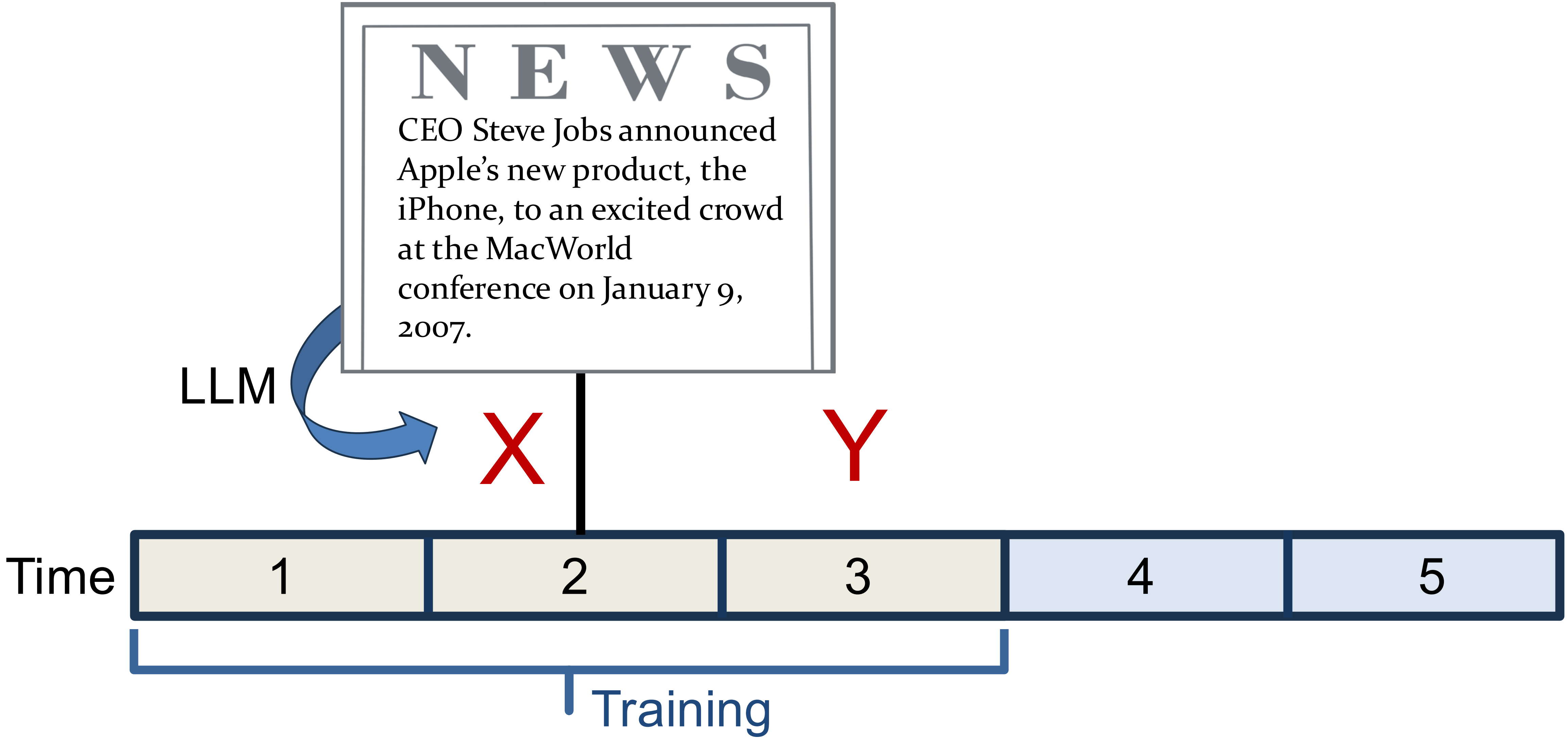
Disadvantages

- Models we create will never be as powerful as frontier ones
  - Sarkar (2024) models are not instruction fine-tuned, require additional training/regression

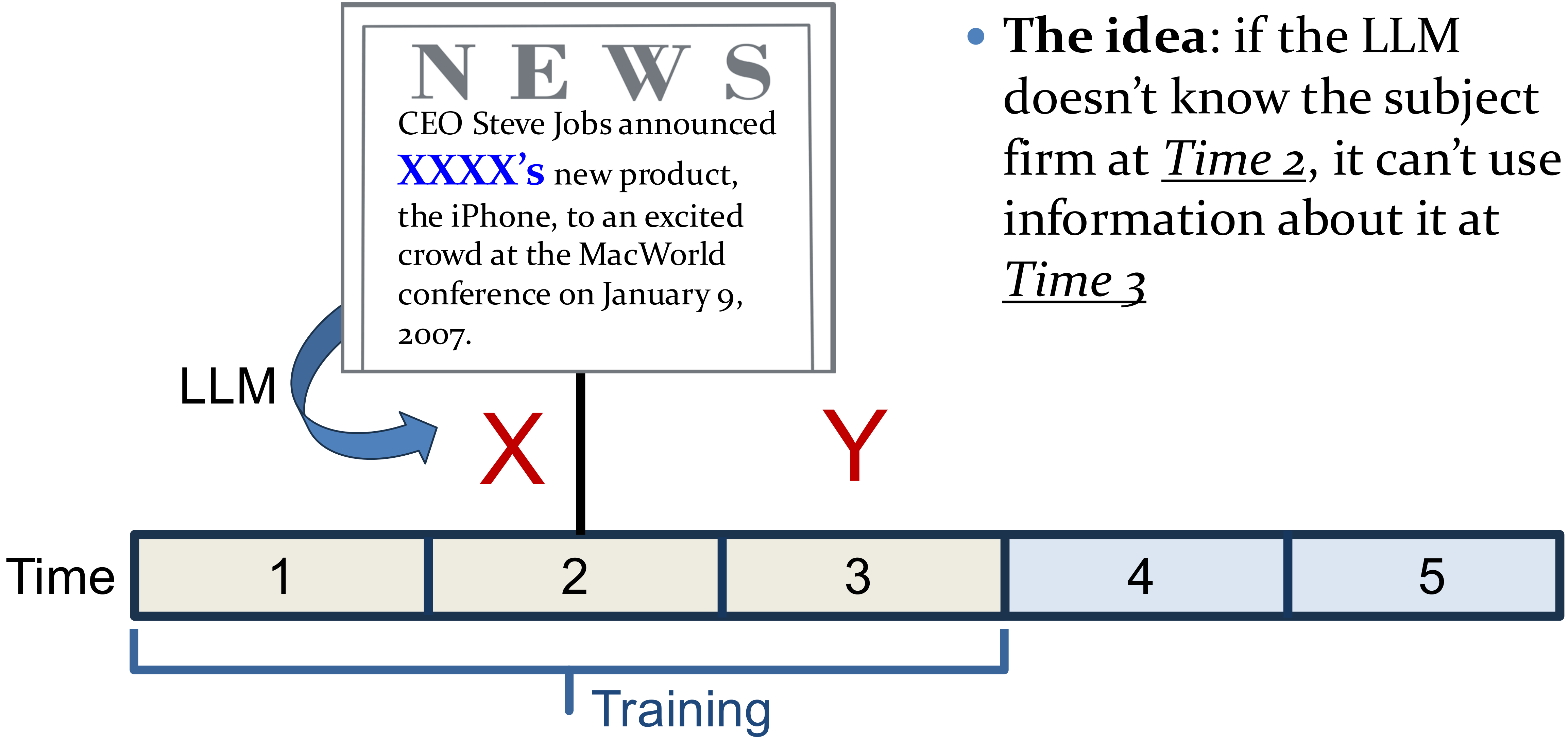
Using frontier LLMs after training cutoff – limited use-cases

- Available time window is shrinking, and some are already real-time

Solution #2: Masking
(Changing the text)



Solution #2: Masking



- **The idea**: if the LLM doesn't know the subject firm at *Time 2*, it can't use information about it at *Time 3*

Solution #2: Masking

- **Little evidence of success** with Masking so far

- For example, Sarkar and Vafa (2024): after masking firm name, GPT-4 can still identify the firm 70% of the time from the other clues in earnings calls

This Paper

- Takes the masking approach, but achieves recognition rates far below what's in the literature (random chance vs. 70%)

- **The Idea**: ask the LLM to use its judgement in removing all identifying information, a process we call **entity neutering**
  - Specifically, we task the LLM with masking and paraphrasing iteratively until the subject firm is unrecognizable by the LLM

- Follows a growing computer science literature called "adversarial anonymization"

Original text:

Coca-Cola (KO) hasn't been a popular stock of late because of weak **soda** sales (volume logged its first quarterly drop in **15 years** in **1Q**). But **RBC** says forthcoming **2Q** results could be the strongest in a while amid a later **Easter** versus **2013**, a **World Cup boost** and the "**Share A Coke**" campaign in which consumers personalize **Coke cans** with names. **RBC** is forecasting **4.4%** volume growth, including **3%** in the **US**, up from **2%** globally in **1Q** and **1%** in **4Q**.

Firm Masked:

Entity_A (*Ticker_A*) hasn't been a popular stock of late because of weak **soda** sales (volume logged its first quarterly drop in **15 years** in **1Q**). But **RBC** says forthcoming **2Q** results could be the strongest in a while amid a later **Easter** versus **2013**, a **World Cup boost** and the "**Share A** *Entity_A* " campaign in which consumers personalize *Entity_A* **cans** with names. **RBC** is forecasting **4.4%** volume growth, including **3%** in the **US**, up from **2%** globally in **1Q** and **1%** in **4Q**.

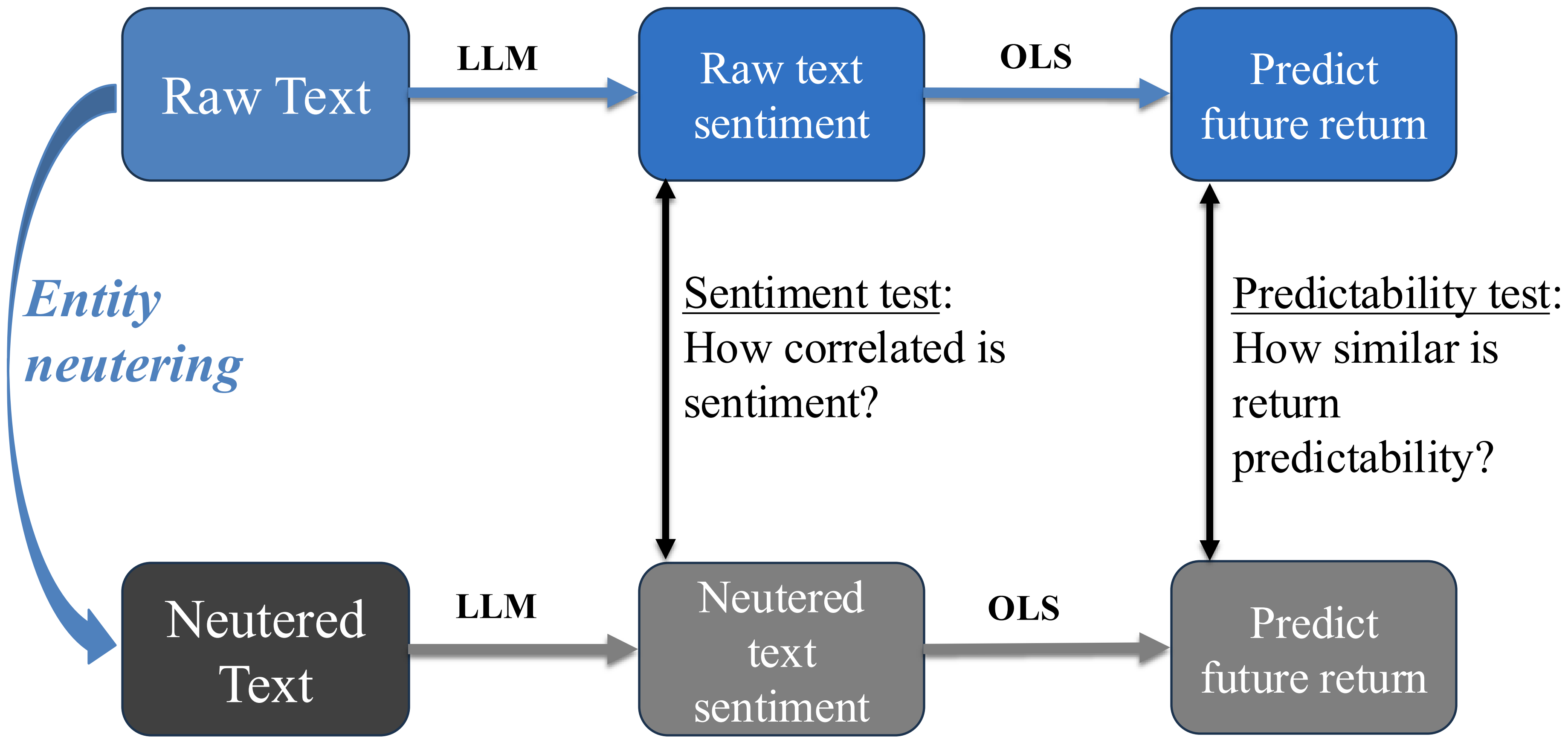
Entity neutered:

Entity_A's most recent performance has failed to gain momentum due to soft demand for *sku_alpha*, having their first fall in demand in *a_years*. Nevertheless, *Entity_B* indicates the upcoming *period_z* could surpass the *previous_time_performance*, helped by an *annual_event*, a *marketing_promo* uptick, and a program allowing users to customize *sku_beta* with a *tag_id*. *Entity_B* expects *x_volume_growth*, including pockets of growth across *region_x* and *region_y*.

What we do

- We take 250,000 news articles from the Dow Jones Archive, 2000-2022
- We ***entity neuter*** each article using GPT-5 nano (“ChatGPT”)
- Then we feed the neutered articles back to ChatGPT and other LLMs to see if they can identify the subject firm of each article
    - We drop articles that can still be identified

What we do



- Next, we extract a sentiment signal from both the raw and neutered text
- Compare the sentiment of the raw vs. neutered text
- Compare the return predictability of the raw vs. neutered text
- Construct a bound on the return predictability coming from look-ahead bias

What we find

Identification rate

- LLMs struggle to identify the subject firm of neutered articles
  - ChatGPT correctly identifies the subject firm **~0.1%** of the time and the date of the articles 0.003% of the time (date +/- 7 days)
  - Llama and Gemma perform even worse (**~0.06%** firm recognition)

Correlation of Sentiment

- Strong correlation of sentiment signals extracted from raw and neutered text: **same sentiment 96% of the time**

What we find

Return predictability

- Binary sentiment (positive and negative) has similar return predictability when comparing raw vs. neutered
  - Neutered a little weaker

- Stronger differences when we condition on sentiment magnitude (e.g., strong positive and strong negative)

- By comparing the improvement in MSE of raw vs. neutered sentiment when predicting returns, we bound look-ahead bias to a range of between 25% and 43%

Data

The Dow Jones Archive

- We begin with all articles that cross the Dow Jones Newswires between 2000 and 2022
  - Mostly wire stories but also includes WSJ articles, Barron's and firm press releases

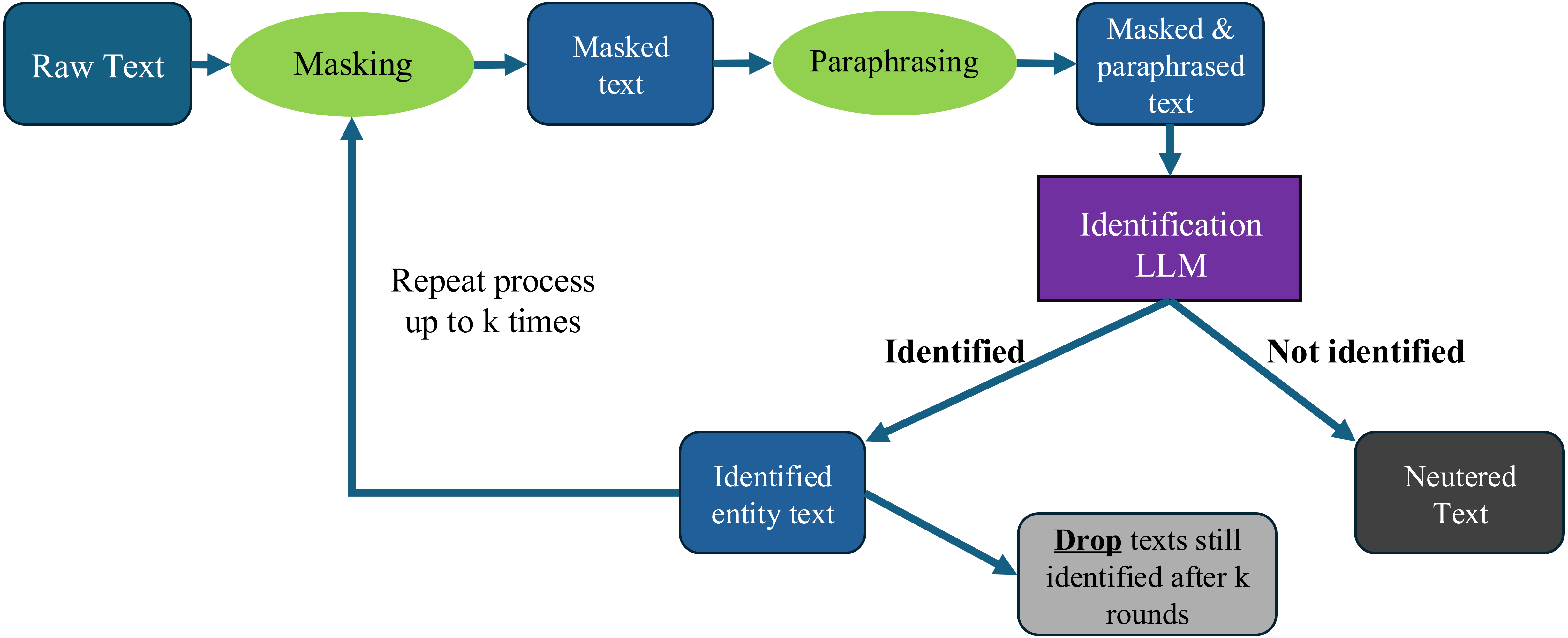
- We follow the filtering approach of Ke, Kelly and Xiu (2019) and then take a random sample of 250,000 articles.
  - Sample = articles of non-trivial length, about a single firm, which can be matched to a CRSP ticker
  - Why a random sample and not all articles? $$

Why the Dow Jones Archive?

- To investigate how much text-based predictability is coming from look-ahead bias, we need to start with text that has a chance of predicting returns
- Starting with Tetlock, Saar-Tsechansky and Macskassy (2008), many papers have found predictability from the Dow Jones Archive
- Most find textual sentiment extracted from day t articles predicts returns over the next few days (t+1 – t+5)
- Stronger predictability from negative sentiment

The *Entity Neutering procedure*

The process


Raw Text
Masking
Masked text
Paraphrasing
Masked & paraphrased text
Identification LLM
Identified
Not identified
Repeat process up to k times
Identified entity text
Neutered Text
Drop texts still identified after k rounds

The **masking** prompt

- We feed each article to GPT-5 nano via the API with the following prompt:

  Your role is to ANONYMIZE all text that is provided by the user. The text you need to anonymize will be about firm {NAME}, its ticker, {TICKER}, and its industry, {INDUSTRY}. Any instance of the firm, their products, the industry or industry related areas they are involved in should be anonymized.

The **masking** prompt

- We give some specifics:

  Also ANONYMIZE any other identifiers related to companies including the names of individuals, locations, industries, sectors, product names and types and generic product lines, services, times, years, dates and all numbers and percentages in the text, including units.

  These should be replaced with name_x, location_x, …

The **masking** prompt

- But we leave it open-ended and up to the LLM's "judgement"

You should also ANONYMIZE any other information which one could use to identify the company or to make an educated guess about its identity…

After you have anonymized a text, NOBODY, not even an expert financial analyst should be able to use the text to identify the company or the industry the company operates in.

The **paraphrasing** prompt

You are to paraphrase text ensuring that its core information content is unchanged, such that someone who has memorized the text could not recognize it.

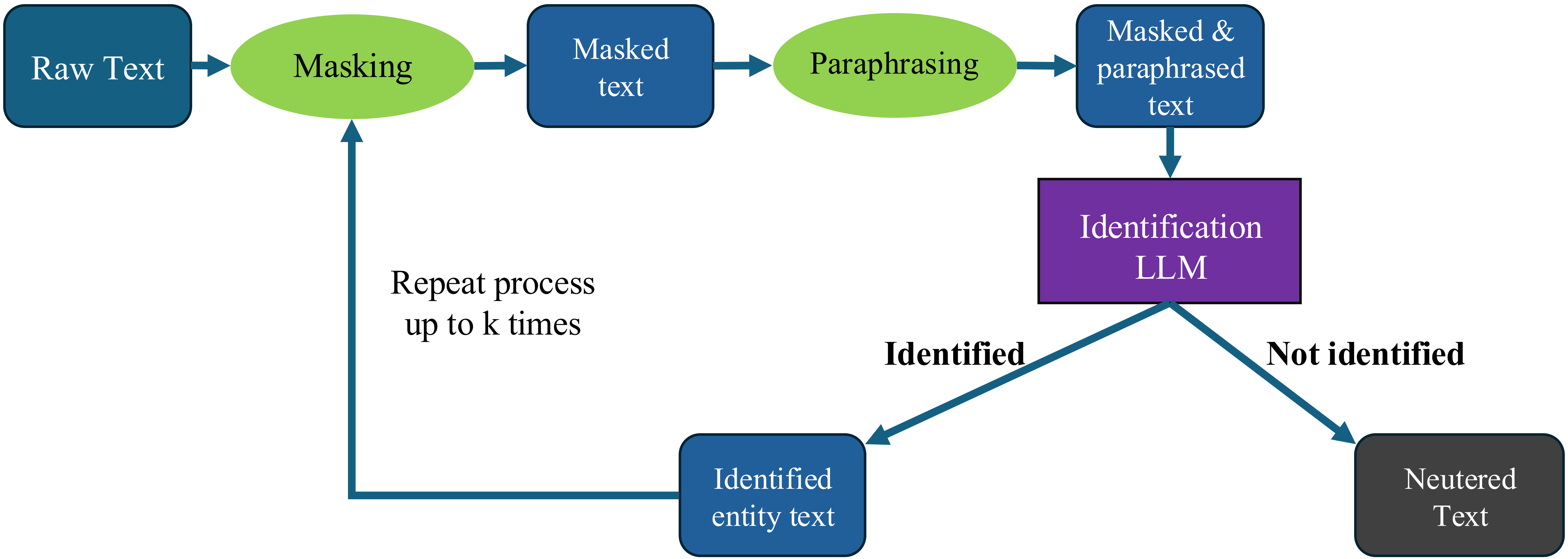
Make sure to replace all unique or identifying words and phrases. Also change the sentence structure so that the resulting text is completely different from the original. This includes changing quotes and other highly recognizable structures in the text.

The text provided will be about firm {NAME}, its ticker, {TICKER}, and its industry, {INDUSTRY}.

Can the LLM recognize the firm in the neutered text?

The **identification** test

- We feed the neutered text back to a new instance of ChatGPT (or another LLM) and ask it to identify the subject firm



How often does ChatGPT correctly guess the firm name?

Firm	Model	# parameters	identification rate	# obs.
OpenAI	GPT-5 nano	unknown	$\boxed{0.09\%}$	255,328

- ChatGPT correctly guesses the firm name **0.09%** of the time (241 texts)
- Benchmarks:
  - Random guessing = 0.1%
  - Most popular firm (Microsoft) = 0.15%

How do other LLMs do?

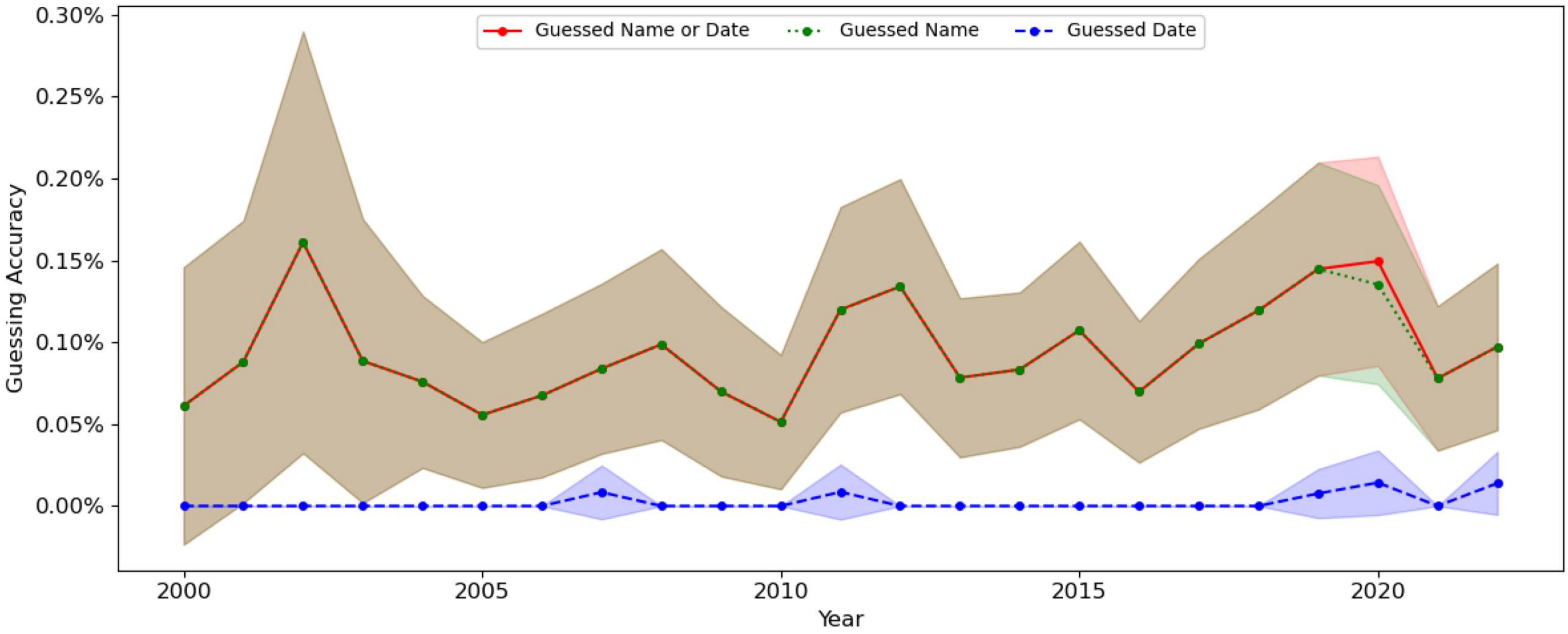
Firm	Model	# parameters	identification rate	# obs.
OpenAI	GPT-5 nano	unknown	0.09%	255,328
Meta	Llama 3.2	3B	0.03%	50,000
Google	Gemma 3	4B	0.06%	50,000

- On random subsamples, **other LLMs** guess the name correctly **0.03%** and **0.06%** of the time

Is the identification rate low because of the sample period?

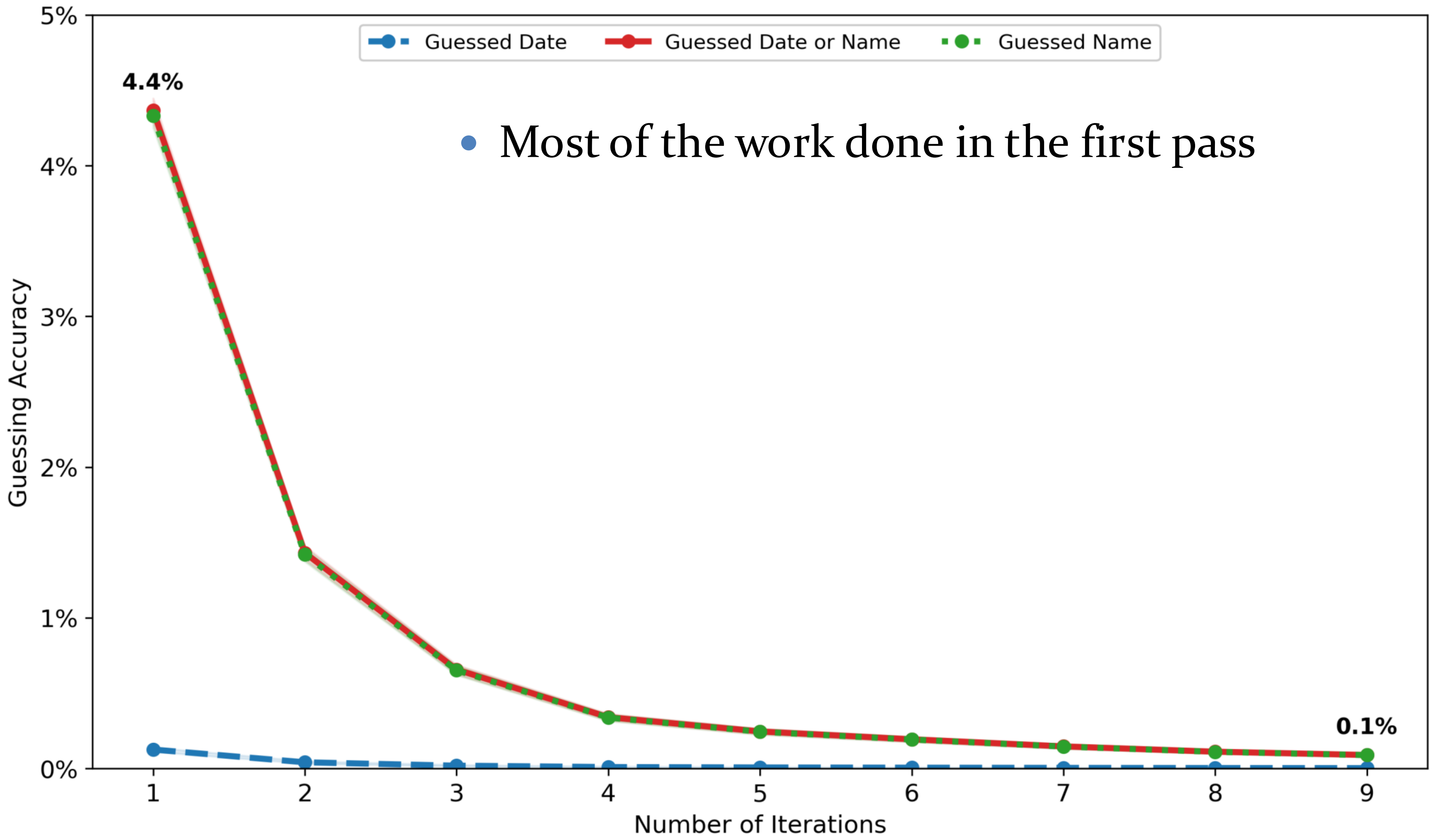
- LLMs are likely trained on *more recent data* and so may better identify more recent firms
- Next slide: We plot firm and date identification rates by year

Guessing accuracy over time



- Firm name (green) and Date (blue) identification rates are both largely flat

How the identification rate changes by N of iterations

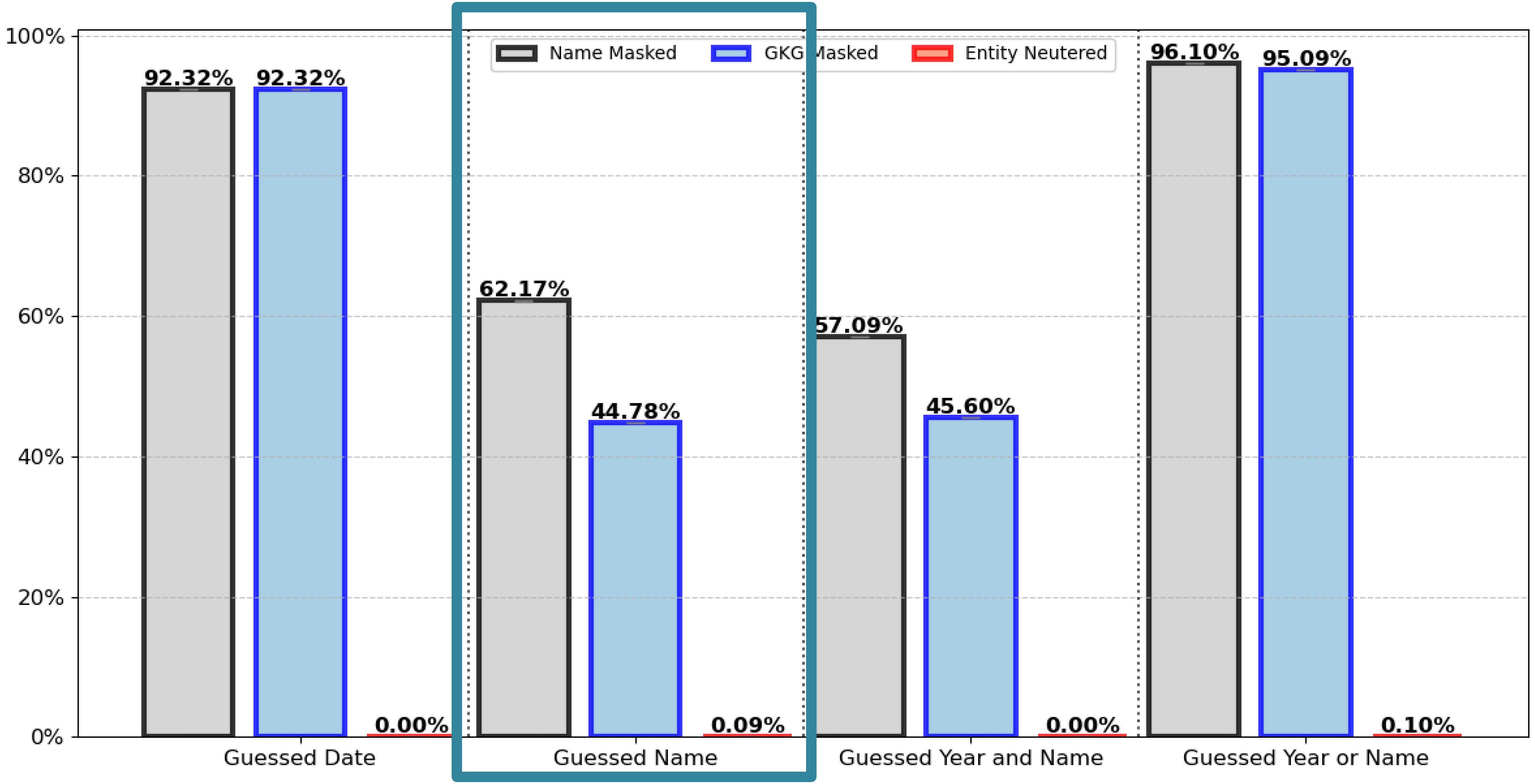


- Most of the work done in the first pass

Comparing entity neutering vs. simple masking

- Sarkar and Vafa (2024) mask firm names
- Glasserman and Lin (2023) mask firm names + 20 related terms in the Google Knowledge Graph (GKG)

Method comparison for ***name***



How correlated is the sentiment extracted from raw vs. neutered text?

The sentiment test

- Entity neutering appears effective but we may have “thrown the baby out with the bathwater”
- By asking the LLM to mask far more than firm name, we may have lost valuable content
- We take neutered articles and ask the LLM to extract sentiment from both the raw and neutered versions of the text

The sentiment prompt

- For the raw and neutered text of each article we prompt GPT with:

  “You are an expert Financial Analyst. Your task is to read news articles and ***infer if the reported news is a good or bad signal*** *for the future returns of the security being written about.* ***Based on your answer, I will choose to buy or sell that security.*** *After reading the news, provide a bull or bear signal as two numbers: the direction and the magnitude…”*

Sentiment extracted from raw & neutered articles is highly correlated

- When sentiment is detected in both the raw and neutered versions of an article, they share the same sentiment
  - **96%** of the time when positive
  - **97%** when negative

How similar is the **return predictability** from the raw vs. neutered sentiment?

Comparing return predictability

- Now we compare how sentiment from the raw vs. neutered articles predicts returns in event-time

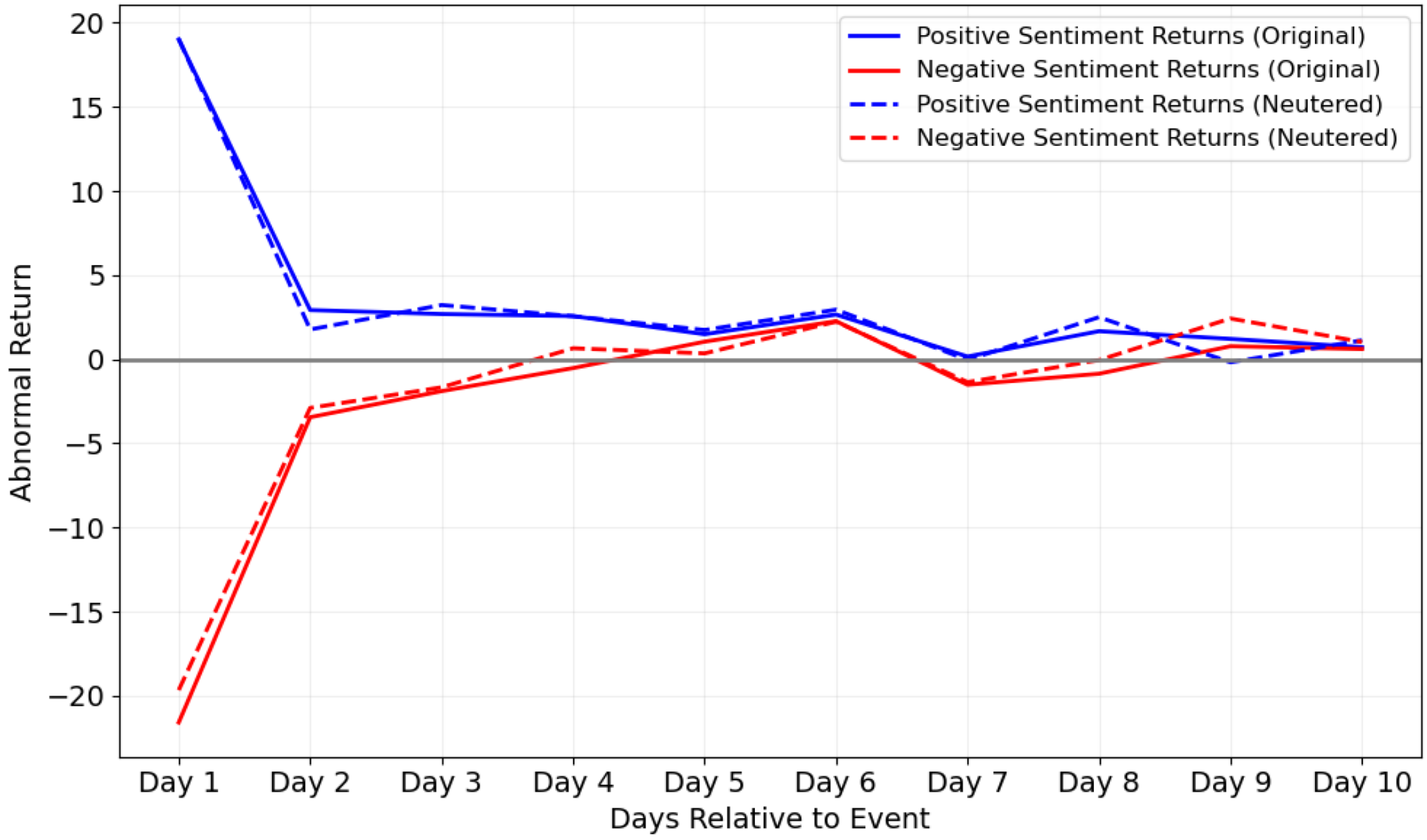
- For each article, we compute Daniel, Grinblatt, Titman and Wermers (DGTW) abnormal returns

- Event-time pictures in the next slide for the raw & neutered, as well as positive & negative sentiment (4 pictures total)

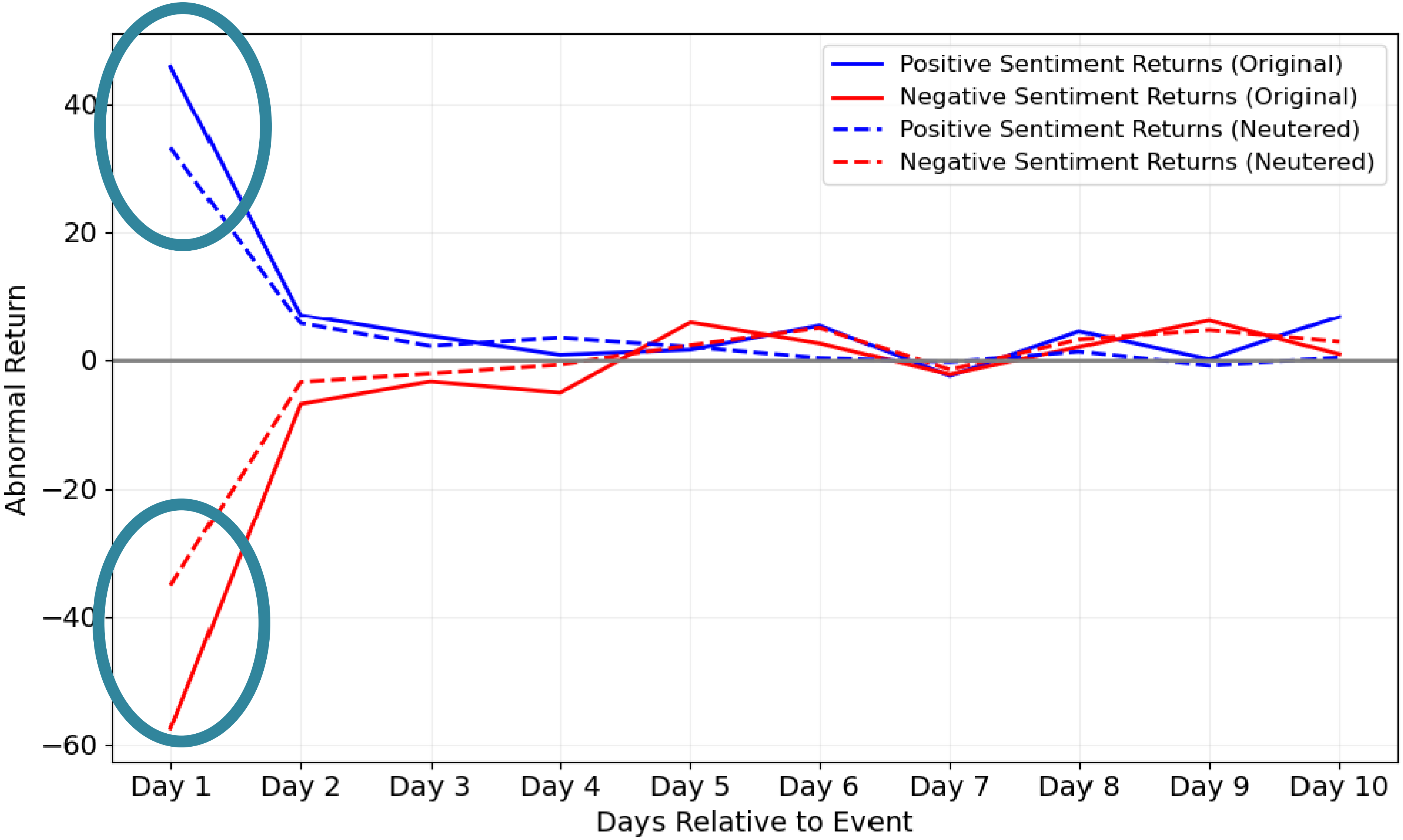
Returns in event time for Positive & Negative sentiment articles

- Plots average DGTW returns (bps) in event-time by binary positive/negative classification

- Overall, similar predictability raw vs. neutered



Returns conditioning on strong sentiment



Strong sentiment =
1 sd above the mean

The event-time regression analogue

- What follows is a panel regression with controls analogous to Tetlock et al. (2008)
  - Controls include day *t* DGTW return, historical DGTW returns and share turnover
- Odd columns use the raw text, even ones use the neutered text
- First four columns predict day *t+1* return, last four predict day *t+2* through *t+5*

Return predictability for day *t+1*

Approximately similar coefficients when using binary sentiment:

- **15 bps** & **-28 bps** for positive & negative among **raw**
- **18 bps** & **-23 bps** among **neutered**

	Outcome: $DGTW_{i,t+1}$				$DGTW_{i,(t+2) \rightarrow (t+5)}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Raw	Neutered	Raw	Neutered	Raw	Neutered	Raw	Neutered
$\mathbb{1}$ Positive sentiment$_{i,t}$	14.46***	17.56***			3.07	5.33*		
	[7.38]	[9.69]			[0.80]	[1.69]		
$\mathbb{1}$ Negative sentiment$_{i,t}$	-28.29***	-22.99***			-9.77**	-5.76*		
	[-13.59]	[-11.90]			[-2.56]	[-1.82]		
$\mathbb{1}$ Strong positive$_{i,t}$			43.25***	31.91***			11.54*	13.17**
			[10.42]	[8.42]			[1.71]	[2.25]
$\mathbb{1}$ Weak positive$_{i,t}$			9.97***	15.13***			1.75	4.03
			[5.10]	[8.12]			[0.45]	[1.23]
$\mathbb{1}$ Weak negative$_{i,t}$			-18.95***	-17.61***			-8.79**	-5.75*
			[-9.37]	[-8.74]			[-2.27]	[-1.74]
$\mathbb{1}$ Strong negative$_{i,t}$			-65.58***	-39.24***			-13.57**	-5.67
			[-15.32]	[-11.82]			[-2.15]	[-1.08]
# observations	242,234	242,234	242,234	242,234	242,145	242,145	242,145	242,145
# clusters (date)	5,781	5,781	5,781	5,781	5,781	5,781	5,781	5,781
Adj. R^2	0.006	0.005	0.007	0.005	0.002	0.002	0.002	0.002

Return predictability for day *t+1*

Larger spreads when conditioning on magnitude:

- **43 bps** & **-66 bps** for strong positive & strong negative among **raw**
- **31 bps** & **-39 bps** among **neutered**

	Outcome: $DGTW_{i,t+1}$				$DGTW_{i,(t+2) \to (t+5)}$			
	(1) Raw	(2) Neutered	(3) Raw	(4) Neutered	(5) Raw	(6) Neutered	(7) Raw	(8) Neutered
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Trading strategy returns: results similar to event-time

- **Calendar-time trading strategy**:
  - long positive sentiment firms
  - short negative sentiment ones & rebalance daily

- **Similar** alpha when trading based on raw vs. neutered binary sentiment

- No loadings on FF6 factors → look-ahead bias unlikely to enter via factor characteristics

Dep. var.: Long short$_{t+1}$

	(1) Raw	(2) Neutered	(3) Difference	(4) Raw & strong	(5) Neutered & strong	(6) Difference & strong
α	48.34***	43.14***	5.20**	70.09***	49.83***	20.26***
	[18.18]	[15.89]	[2.50]	[16.01]	[14.55]	[5.28]
Market	-0.04	-0.04	-0.00	-0.02	-0.04	0.02
	[-1.29]	[-1.18]	[-0.08]	[-0.46]	[-0.89]	[0.36]
HML	0.12*	0.05	0.07	0.15	0.09	0.06
	[1.65]	[0.70]	[1.21]	[1.21]	[1.22]	[0.54]
SMB	0.15**	0.13**	0.02	0.09	0.15**	-0.06
	[2.35]	[1.97]	[0.43]	[0.92]	[1.99]	[-0.61]
UMD	0.04	0.00	0.04	0.10*	0.05	0.05
	[1.11]	[0.03]	[1.46]	[1.71]	[1.11]	[1.03]
RMW	0.02	-0.04	0.06	-0.14	-0.02	-0.12
	[0.25]	[-0.51]	[0.96]	[-1.12]	[-0.18]	[-1.14]
CMA	-0.17	-0.01	-0.16*	-0.17	0.01	-0.19
	[-1.44]	[-0.06]	[-1.77]	[-1.04]	[0.10]	[-1.17]
# observations	5,702	5,702	5,702	5,702	5,702	5,702
Adj. R^2	0.003	0.001	0.001	0.001	0.002	0.000

Calculating a **bound** on look-ahead bias

- Assume masking & paraphrasing only reduces (never improves) the value-relevant return information the LLM extracts from text

- Then there are 2 reasons why the return predictability for raw is greater than that for neutered:
  (1) Look-ahead bias
  (2) Better information in the raw text

- Thus, we can view the difference between the two as an *upper bound on look-ahead bias*

Calculating a **bound** on look-ahead bias

To calculate the bound:

- Take the baseline Tetlock et al. (2008) specifications for predicting future returns and
- Compare the improvement in MSE when adding raw sentiment variables vs. neutered ones

Bound calculation for return predictability

Compared to a specification with only controls,

- **adding the raw sentiment** variables in column (1) **decreases MSE by 374**

	Outcome: $DGTW_{i,t+1}$				$DGTW_{i,(t+2) \to (t+5)}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
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# clusters (date)	5,781	5,781	5,781	5,781	5,781	5,781	5,781	5,781
Adj. R^2	0.006	0.005	0.007	0.005	0.002	0.002	0.002	0.002

Bound calculation for return predictability

Compared to a specification with only controls,

- adding the **neutered sentiment** variables in column (2) **decreases MSE by 281**

	Outcome: $\text{DGTW}_{i,t+1}$				$\text{DGTW}_{i,(t+2)\to(t+5)}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Raw	Neutered	Raw	Neutered	Raw	Neutered	Raw	Neutered
$\mathbb{1}$ Positive $\text{sentiment}_{i,t}$	14.46***	17.56***			3.07	5.33*		
	[7.38]	[9.69]			[0.80]	[1.69]		
$\mathbb{1}$ Negative $\text{sentiment}_{i,t}$	-28.29***	-22.99***			-9.77**	-5.76*		
	[-13.59]	[-11.90]			[-2.56]	[-1.82]		
$\mathbb{1}$ Strong $\text{positive}_{i,t}$			43.25***	31.91***			11.54*	13.17**
			[10.42]	[8.42]			[1.71]	[2.25]
$\mathbb{1}$ Weak $\text{positive}_{i,t}$			9.97***	15.13***			1.75	4.03
			[5.10]	[8.12]			[0.45]	[1.23]
$\mathbb{1}$ Weak $\text{negative}_{i,t}$			-18.95***	-17.61***			-8.79**	-5.75*
			[-9.37]	[-8.74]			[-2.27]	[-1.74]
$\mathbb{1}$ Strong $\text{negative}_{i,t}$			-65.58***	-39.24***			-13.57**	-5.67
			[-15.32]	[-11.82]			[-2.15]	[-1.08]
# observations	242,234	242,234	242,234	242,234	242,145	242,145	242,145	242,145
# clusters (date)	5,781	5,781	5,781	5,781	5,781	5,781	5,781	5,781
Adj. R^2	0.006	0.005	0.007	0.005	0.002	0.002	0.002	0.002

Bound calculation for return predictability

Comparing MSE improvements in (1) and (2):

- **At most** (374-281)/374 = **25%** is coming from look-ahead bias

	Outcome: $\text{DGTW}_{i,t+1}$				$\text{DGTW}_{i,(t+2)\rightarrow(t+5)}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Raw	Neutered	Raw	Neutered	Raw	Neutered	Raw	Neutered
$\mathbb{1}$ Positive $\text{sentiment}_{i,t}$	14.46***	17.56***			3.07	5.33*		
	[7.38]	[9.69]			[0.80]	[1.69]		
$\mathbb{1}$ Negative $\text{sentiment}_{i,t}$	-28.29***	-22.99***			-9.77**	-5.76*		
	[-13.59]	[-11.90]			[-2.56]	[-1.82]		
$\mathbb{1}$ Strong $\text{positive}_{i,t}$			43.25***	31.91***			11.54*	13.17**
			[10.42]	[8.42]			[1.71]	[2.25]
$\mathbb{1}$ Weak $\text{positive}_{i,t}$			9.97***	15.13***			1.75	4.03
			[5.10]	[8.12]			[0.45]	[1.23]
$\mathbb{1}$ Weak $\text{negative}_{i,t}$			-18.95***	-17.61***			-8.79**	-5.75*
			[-9.37]	[-8.74]			[-2.27]	[-1.74]
$\mathbb{1}$ Strong $\text{negative}_{i,t}$			-65.58***	-39.24***			-13.57**	-5.67
			[-15.32]	[-11.82]			[-2.15]	[-1.08]
# observations	242,234	242,234	242,234	242,234	242,145	242,145	242,145	242,145
# clusters (date)	5,781	5,781	5,781	5,781	5,781	5,781	5,781	5,781
Adj. R^2	0.006	0.005	0.007	0.005	0.002	0.002	0.002	0.002

Bound calculation for return predictability

Comparing the MSE improvements in (3) and (4):

- **At most 43%** is coming from look-ahead bias

	Outcome: $DGTW_{i,t+1}$				$DGTW_{i,(t+2) \rightarrow (t+5)}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Raw	Neutered	Raw	Neutered	Raw	Neutered	Raw	Neutered
$\mathbb{1}$ Positive sentiment$_{i,t}$	14.46***	17.56***			3.07	5.33*		
	[7.38]	[9.69]			[0.80]	[1.69]		
$\mathbb{1}$ Negative sentiment$_{i,t}$	-28.29***	-22.99***			-9.77**	-5.76*		
	[-13.59]	[-11.90]			[-2.56]	[-1.82]		
$\mathbb{1}$ Strong positive$_{i,t}$			43.25***	31.91***			11.54*	13.17**
			[10.42]	[8.42]			[1.71]	[2.25]
$\mathbb{1}$ Weak positive$_{i,t}$			9.97***	15.13***			1.75	4.03
			[5.10]	[8.12]			[0.45]	[1.23]
$\mathbb{1}$ Weak negative$_{i,t}$			-18.95***	-17.61***			-8.79**	-5.75*
			[-9.37]	[-8.74]			[-2.27]	[-1.74]
$\mathbb{1}$ Strong negative$_{i,t}$			-65.58***	-39.24***			-13.57**	-5.67
			[-15.32]	[-11.82]			[-2.15]	[-1.08]
# observations	242,234	242,234	242,234	242,234	242,145	242,145	242,145	242,145
# clusters (date)	5,781	5,781	5,781	5,781	5,781	5,781	5,781	5,781
Adj. R^2	0.006	0.005	0.007	0.005	0.002	0.002	0.002	0.002

Bound calculation for return predictability

- For predicting t+2 through t+5, we compute a bound **of 39% using columns 5 and 6**
- and **39% using columns 7 and 8**

	Outcome: $DGTW_{i,t+1}$				$DGTW_{i,(t+2)\rightarrow(t+5)}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Raw	Neutered	Raw	Neutered	Raw	Neutered	Raw	Neutered
$\mathbb{1}$ Positive sentiment$_{i,t}$	14.46***	17.56***			3.07	5.33*		
	[7.38]	[9.69]			[0.80]	[1.69]		
$\mathbb{1}$ Negative sentiment$_{i,t}$	-28.29***	-22.99***			-9.77**	-5.76*		
	[-13.59]	[-11.90]			[-2.56]	[-1.82]		
$\mathbb{1}$ Strong positive$_{i,t}$			43.25***	31.91***			11.54*	13.17**
			[10.42]	[8.42]			[1.71]	[2.25]
$\mathbb{1}$ Weak positive$_{i,t}$			9.97***	15.13***			1.75	4.03
			[5.10]	[8.12]			[0.45]	[1.23]
$\mathbb{1}$ Weak negative$_{i,t}$			-18.95***	-17.61***			-8.79**	-5.75*
			[-9.37]	[-8.74]			[-2.27]	[-1.74]
$\mathbb{1}$ Strong negative$_{i,t}$			-65.58***	-39.24***			-13.57**	-5.67
			[-15.32]	[-11.82]			[-2.15]	[-1.08]
# observations	242,234	242,234	242,234	242,234	242,145	242,145	242,145	242,145
# clusters (date)	5,781	5,781	5,781	5,781	5,781	5,781	5,781	5,781
Adj. R^2	0.006	0.005	0.007	0.005	0.002	0.002	0.002	0.002

Conclusion

- We propose a new way to address the look-ahead bias problem by **tasking the LLM** with de-identification

- This approach achieves much **lower entity recognition rates** than other masking methods, and largely **preserves predictive power** in our setting

- Where there are differences in predictability between raw and neutered we show **how a bound on look-ahead bias** can be estimated using this approach